

1.Code for Struct Account:

```
#include<bits/stdc++.h>
```

```
using namespace std;
```

```
struct Account {
```

```
    string name;
```

```
    int accountNumber;
```

```
    double balance;
```

```
    Account(string n, int accNum, double bal) : name(n), accountNumber(accNum),  
    balance(bal) {}
```

```
    void deposit(double amount) {
```

```
        balance += amount;
```

```
    }
```

```
    void withdraw(double amount) {
```

```
        if (amount <= balance) {
```

```
            balance -= amount;
```

```
        } else {
```

```
            cout << "Insufficient funds" << endl;
```

```
        }
```

```
    }
```

```
    void display() const {
```

```
        cout << "Account Name: " << name << endl;
```

```

        cout << "Account Number: " << accountNumber << endl;

        cout << "Balance: $" << balance << endl;
    }
};

int main() {
    Account acc("John Doe", 123456, 1000.0);

    acc.display();

    acc.deposit(500.0);

    acc.display();

    acc.withdraw(200.0);

    acc.display();

    acc.withdraw(1500.0); // Attempt to withdraw more than balance

    return 0;
}

```

2.Code for Class Account:

```

#include<bits/stdc++.h>

using namespace std;

class Account {

    private:

        string accountNumber;

        string accountHolderName;

        double balance;

    public:

```

```
Account(string accNum, string accHolderName, double initialBalance) {  
    accountNumber = accNum;  
    accountHolderName = accHolderName;  
    balance = initialBalance;  
}
```

```
void deposit(double amount) {  
    if (amount > 0) {  
        balance += amount;  
        cout << "Deposited: " << amount << endl;  
    } else {  
        cout << "Invalid deposit amount." << endl;  
    }  
}
```

```
void withdraw(double amount) {  
    if (amount > 0 && amount <= balance) {  
        balance -= amount;  
        cout << "Withdrew: " << amount << endl;  
    } else {  
        cout << "Invalid withdrawal amount." << endl;  
    }  
}
```

```
void displayAccountInfo() {  
    cout << "Account Number: " << accountNumber << endl;
```

```

        cout << "Account Holder: " << accountHolderName << endl;

        cout << "Balance: " << balance << endl;
    }
};

int main() {
    Account myAccount("123456", "Muhammad Sumon", 1000.0);

    myAccount.displayAccountInfo();

    myAccount.deposit(500.0);

    myAccount.displayAccountInfo();

    myAccount.withdraw(200.0);

    myAccount.displayAccountInfo();

    return 0;
}

```

3.Code for a Library book record:

```

#include <bits/stdc++.h>

using namespace std;

class Book {
    private:

        string title;

        string author;

        int publicationYear;

    public:

        Book(string t, string a, int p) : title(t), author(a), publicationYear(p) {}

        void setTitle(string t) {

```

```

        title = t;
    }
    void setAuthor(string a) {
        author = a;
    }
    void setPublicationYear(int p) {
        publicationYear = p;
    }

    void displayInfo() {
        cout << "Title: " << title << endl;
        cout << "Author: " << author << endl;
        cout << "Publication Year: " << publicationYear << endl;
    }
};

int main() {
    Book book1("The kite Runner", "Khaled Hosseini", 2003);
    book1.displayInfo();

    return 0;
}

```

In the old C version, any function could directly reach into a book's struct and leave title garbage/uninitialized, since the fields had no protection. Now that title, author, and publication year are private, no outside code can touch them directly — every change has to go through setData(), so that kind of accidental, unchecked corruption of a book's data is no longer possible.

4.Code for Class Student:

```
#include<bits/stdc++.h>
```

```
using namespace std;
```

```
class Student {
```

```
private:
```

```
    int rollNumber;
```

```
    string name;
```

```
    double marks;
```

```
public:
```

```
    Student(string n, int r, double m) : name(n), rollNumber(r), marks(m) {}
```

```
    void setName(string n) {
```

```
        name = n;
```

```
    }
```

```
    void setRollNumber(int r) {
```

```
        rollNumber = r;
```

```
    }
```

```
    void setMarks(double m) {
```

```
        marks = m;
```

```
    }
```

```
    char getGrade() {
```

```
        if (marks >= 90) {
```

```
            return 'A';
```

```
        } else if (marks >= 80) {
```

```
            return 'B';
```

```
    } else if (marks >= 70) {  
        return 'C';  
    } else if (marks >= 60) {  
        return 'D';  
    } else {  
        return 'F';  
    }  
}
```

```
void displayInfo() {  
    cout << "Name: " << name << endl;  
    cout << "Roll Number: " << rollNumber << endl;  
    cout << "Marks: " << marks << endl;  
    cout << "Grade: " << getGrade() << endl;  
}
```

```
};
```

```
int main() {  
    Student student1("Muhammad Sumon",44, 85.5);  
    student1.displayInfo();  
    return 0;  
}
```